

UNIT-I

Quantum Physics

Introduction:

Advances in solid state physics are made by using the principles of Quantum mechanics. The energy of the electron in the solid, black body radiation, Spectra of solids, magnetism, superconductivity, etc., are all explained only by quantum mechanics.

At the end of the 19th century, physicists had every reason to regard the Newtonian laws governing the motion of material bodies and man well less of electromagnetism, as fundamental laws of Physics. They believed that there should be some limitation on the validity of these laws which constitute “classical mechanics” to understand the submicroscopic world of the atom and its constituents, it became necessary to introduce new ideas and concepts which led to the mathematical formulation of “Quantum mechanics” during 1925, that had no immediate and spectacular success in the explanation of the experimental observations. Quantum mechanics provided the key to the understanding of the behavior of very small objects like the atoms and it constitutes the failure of classical mechanics when applied to the submicroscopic world of the atom arose from the possibility the characterizing the instantaneous state of the particle in motion by privies positions and velocities.

In this unit we will study the development of the Quantum mechanics, Black body radiation, Planck's law, photoelectric effect, Compton effect, Broglie hypothesis, Heisenberg's uncertainty principle, Schrödinger's wave equation and its application.

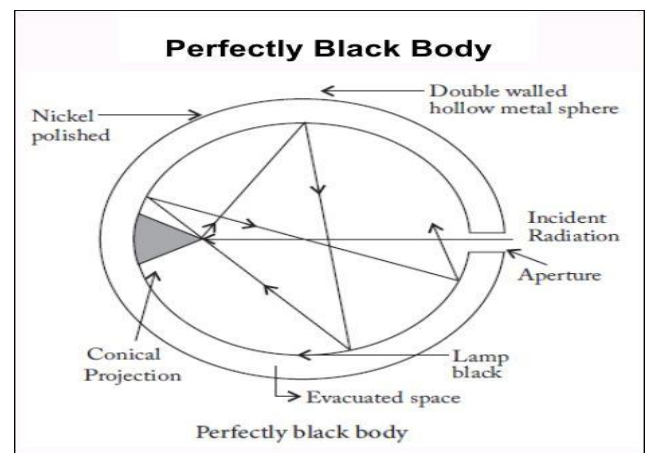
1.1 Black body radiation:

A black body is a theoretical object that absorbs 100% of the radiation that hits it. Therefore it reflects no radiation and appears perfectly black.

Roughly we can say that the stars radiate like blackbody radiators. This is important because it means that we can use the theory for blackbody radiators to infer things about stars.

At a particular temperature the black body would emit the maximum amount of energy possible for that temperature. The radiation emitted by black body is known as black body radiation.

Blackbody radiation does not depend on the type of object emitting it. Entire spectrum of blackbody radiation depends on only one parameter, the temperature, T .



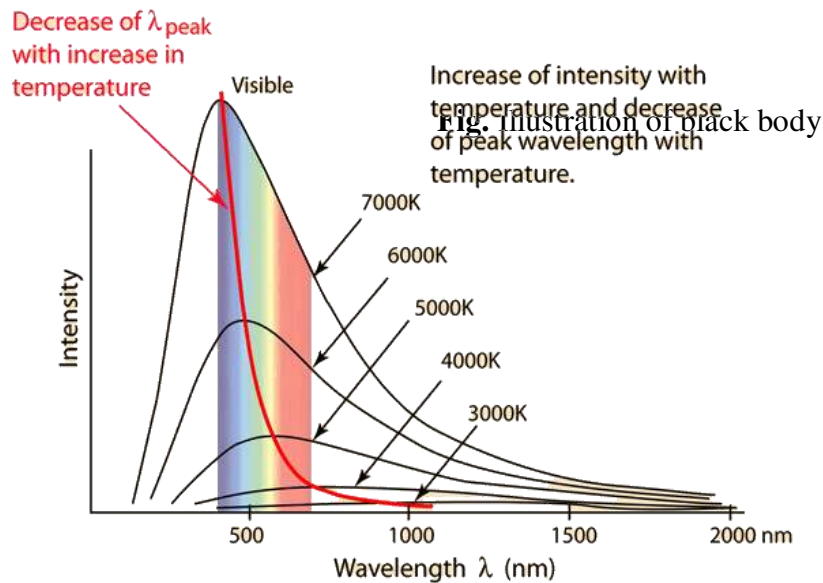


Fig. Energy density Vs wavelength at given temperature

1. At given temperature, the energy is not uniformly distributed in the radiation spectrum of the blackbody.
2. At a given temperature, the intensity of radiation increases with increase in wavelength and at a particular wavelength, its value is maximum. With further increase in wavelength, the intensity of radiation decreases.
3. With the increase in temperature, maximum wavelength (λ_m) decreases, when λ_m is the wavelength at which the maximum emission of energy takes place.
4. There is increase in energy emission with the increase in temperature corresponding to all the wavelengths.
5. The area under each curve the total energy emitted for the complete spectrum at a particular temperature. With the increase in temperature, this area increases.

1.2 Planck's law of black body radiation:

Max Plank, a German physicist derived an equation which successfully accounted for the spectrum of the blackbody radiation. He incorporated a new idea in his deduction of Plank eqn. that the probability of emission of radiation decreases as its frequency increases so that, the curve slopes down in the high frequency region. The oscillators in the blackbody can have only a

discrete set of energy values. Such an assumption was radically different from the basic principles of physics.

The assumption in the derivation of Planck's law is that the wall of the experimental blackbody consists of a very large number of electrical oscillators, with each oscillator vibrating with a frequency of its own. Planck brought two special conditions in his theory. They are

(1) Only an integral multiple of energies $h\nu$ where 'h' is Planck's constant and ' ν ' is frequency of vibration i.e, the allowed energy values are $E = n h \nu$

where $n = 0, 1, 2, \dots$

(2) An oscillator may lose or gain energy by emitting or absorbing radiation of frequency $\nu = (\Delta E / h)$,

where ΔE is the difference in the values of energies of the oscillator before and the emission or absorption had taken place.

Black Body:

A blackbody allows all incident radiation to pass into it (no reflected energy) and internally absorbs all the incident radiation (no energy transmitted through the body). This is true for radiation of all wavelengths and for all angles of incidence. Hence the blackbody is a perfect absorber for all incident radiation. The theory of the energy distribution of blackbody radiation was developed by Planck and first appeared in 1901.

1.3 Planck's Radiation Formula:

Planck assumed the presence of number of resonators (harmonic oscillators) inside the cavity of the black-body and used the quantum hypothesis to derive the formula. According to Planck's quantum hypothesis, an oscillator can have only discrete energies given by

$$E_n = nh\nu = nE$$

$$\text{Where } n = 0, 1, 2, \dots$$

ν is the frequency of the oscillator h is Planck's constant n is an integer (including 0) known as quantum number. Thus an oscillator can have energies 0, $h\nu$, $2h\nu$, And the oscillator cannot emit or absorb energy continuously but in quanta of energy $h\nu$. We shall first calculate average energy of the oscillator of frequency ν . if N is the total number of oscillators and E their total energy, then average energy per Planck's oscillator is given by

$$\overline{E} = E/N$$

Let $N = N_0, N_1, \dots, N_r, \dots$ be the number of oscillators having 0, $h\nu$, $2h\nu$, $rh\nu$, respectively.

The relative probability that an oscillator will have the energy $h\nu$ at temperature T is $\exp(-h\nu/kT)$ where k is Boltzmann constant. In other words, according to Maxwell's law of molecular

Applied Physics

I -B.Tech

motions if ϵ is a certain amount of energy, the probabilities that number of particles with energies 0, ϵ , 2ϵ , $r\epsilon$ are in the ratio

$$1: e^{-\epsilon/kT} : e^{-2\epsilon/kT} : \dots : e^{-r\epsilon/kT}$$

As said earlier if N_0 is the number of resonators having energy zero, then the number of resonators N_1 having energy ϵ will be $N_0 e^{-\epsilon/kT}$, the number of resonators N_2 having energy 2ϵ will be $N_0 e^{-2\epsilon/kT}$ and so on.

Thus, we have

$$N_r = N_0 \exp [-r\epsilon/kT]$$

$$\text{And } N = N_0 + N_1 + N_2 + \dots + N_r + \dots$$

$$= N_0 + N_0 \exp \left[\frac{-\epsilon}{kT} \right] + N_0 \exp \left[\frac{-2\epsilon}{kT} \right] + \dots + N_0 \exp \left[\frac{-r\epsilon}{kT} \right] + \dots$$

$$\text{Put } \exp \left[\frac{-\epsilon}{kT} \right] = x$$

$$N = N_0 (1 + x + x^2 + \dots + x^r)$$

$$N = N_0 (1 - x)^{-1}$$

$$\text{As } (1 - x)^{-1} = 1 + x + x^2 + \dots + x^r$$

Now the total energy E is given by

$$E = N_0 \cdot 0 + \epsilon N_0 \exp[-\epsilon/kT] + 2\epsilon N_0 \exp[-2\epsilon/kT] + \dots + r\epsilon N_0 \exp[-r\epsilon/kT] + \dots$$

$$N_0 \epsilon (1 + 2x + 3x^2 + 4x^3 + \dots + (r+1)x^r)$$

$$E = \frac{N_0 \epsilon}{(1 - x)^2}$$

$$\text{As } 1 + 2x + 3x^2 + \dots + rx^{r-1} + \dots = 1/(1 - x)^2$$

$$\bar{E} = N_0 \epsilon (1 - x)^{-2}$$

Now the average energy of the oscillator is given by

$$\bar{E} = \frac{N_0 \epsilon}{(1 - x)^{-2}} \bigg/ \frac{N_0}{(1 - x)}$$

$$= \frac{\epsilon}{(1 - x)}$$

$$= \frac{h\nu}{1 - e^{-h\nu/kT}} \quad \text{with } E = h\nu$$

$$\bar{E} = \frac{h\nu}{\left\{ \exp\left[\frac{h\nu}{kT}\right] - 1 \right\}}$$

Thus we see that the average energy of oscillator given by Eqn. is different from the energy kT of a classical oscillator.

If we assume that the number of oscillator given by Eqn. $\left[\frac{8\pi\nu^3}{c^3} \right] d\nu$, then we get the total energy per unit volume belonging $d\nu$ or the energy density belonging to the range $d\nu$ as

$$E\nu d\nu = \left[\frac{8\pi\nu^3}{c^3} \right] \left[\frac{h\nu}{\exp\left[\frac{h\nu}{kT}\right] - 1} \right] d\nu$$

This is the famous Planck's radiation law

$$\text{But } \nu = c/\lambda \text{ and } |d\nu| = -\frac{c}{\lambda^2} d\lambda$$

$$\text{ie } E\lambda d\lambda = \frac{8\pi h}{c^3} \left[\frac{c^3}{\lambda^3} \right] \left[\frac{1}{\left\{ \exp\left[\frac{ch}{\lambda kT}\right] - 1 \right\}} \right] \left[\frac{c}{\lambda^2} d\lambda \right]$$

therefore

$$E\lambda d\lambda = \left[\frac{8\pi hc}{\lambda^5} \right] \left[\frac{1}{\exp\left[\frac{ch}{\lambda kT}\right] - 1} \right] d\lambda$$

This equation of Planck well agrees with the experimental value throughout the whole range of wavelength.

With the help of planck's radiation law, wien's displacement law and Rayleigh-Jeans law can be derived in the following way:

For small temperature , λT is small. Now $\exp(hc/\lambda kT) \gg 1$ and 1 can be neglected in the denominator or Eqn.

$$E\lambda d\lambda = \frac{8\pi ch}{\lambda^5} \left\{ \exp\left(\frac{-ch}{\lambda kT}\right) \right\} d\lambda$$

$$= C_1 \lambda^{-5} \left\{ \exp\left(\frac{C_2}{\lambda T}\right) \right\} d\lambda \quad \text{where } C_1 = 8\pi ch, C_2 = \frac{ch}{k}$$

which is **Wein's law**

Rayleigh- Jeans law

For large temperatures λT is large.

$$\text{Now } \exp\left(\frac{ch}{\lambda kT}\right) = 1 + \left(\frac{ch}{\lambda kT}\right) + \frac{1}{2} \left(\frac{ch}{\lambda kT}\right)^2 + \dots$$

$$= 1 + \left(\frac{ch}{\lambda kT}\right)$$

$$E\lambda d\lambda = \frac{8\pi ch}{\lambda^5} \left[\frac{1}{1 + \left(\frac{ch}{\lambda kT}\right) - 1} \right] d\lambda$$

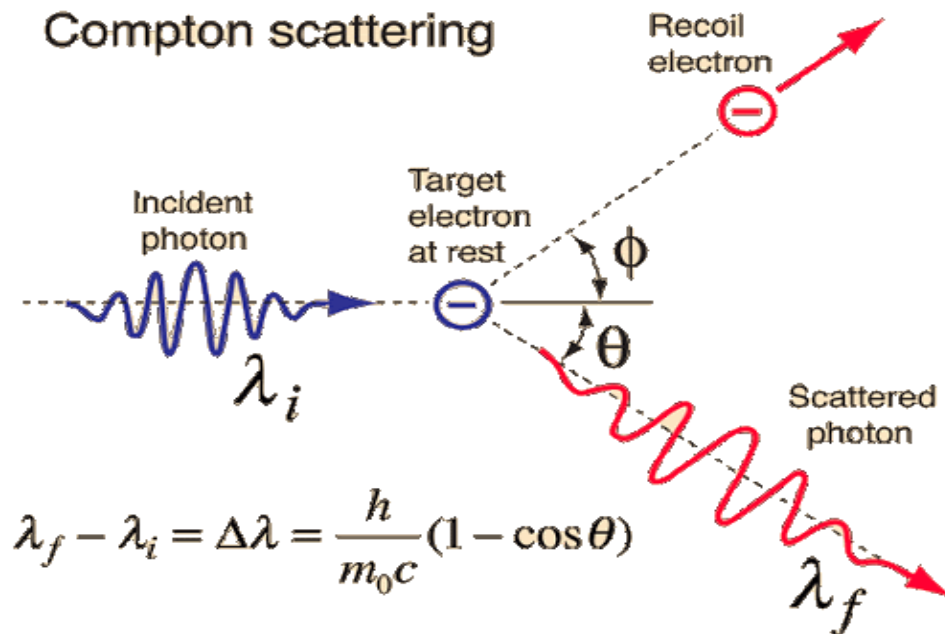
$$\text{i.e. } E\lambda d\lambda = \frac{8\pi ch}{\lambda^5} \times \frac{\lambda kT}{Ch^5} d\lambda = \frac{8\pi KT}{\lambda^4} d\lambda$$

This is **Rayleigh- Jeans law**

where it should be clearly understood that the quantization of energy in the oscillators is not only confined to the atoms of black body but it is a general concept.

1.4 Compton Effect:

Compton effect is the phenomenon in which a collision between a photon and a particle results in an increase in the kinetic energy of the particle and a corresponding increase in the wavelength of the photon. Or An increase in wavelength of X-rays or gamma rays that occurs when they are scattered.



Where λ_i = wavelength of incident radiation

λ_f = wavelength of scattered photon

θ = angle made by the recoiled electron

CASE 1) : When $\theta=0^\circ$

$$\lambda_f - \lambda_i = \Delta\lambda = \frac{h}{m_0 c} (1 - \cos 0^\circ)$$

$$\lambda_f - \lambda_i = \frac{h}{m_0 c} (1 - 1) \text{ since } \cos 0^\circ = 1$$

$$\lambda_f - \lambda_i = \frac{h}{m_0 c} (0)$$

$$\lambda_f - \lambda_i = 0$$

This indicates no scattering along the direction of incidence

CASE 2) : When $\theta=90^\circ$

$$\lambda_f - \lambda_i = \Delta\lambda = \frac{h}{m_0 c} (1 - \cos 90^\circ)$$

$$\lambda_f - \lambda_i = \frac{h}{m_0 c} (1 - 0) \text{ since } \cos 90^\circ = 0$$

$$\lambda_f - \lambda_i = \frac{h}{m_0 c} (1)$$

$$\lambda_f - \lambda_i = \frac{h}{m_0 c} = \frac{6.625 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} = 0.02445 \text{Å}$$

This difference in wavelength is known as Compton wavelength

CASE 3) : When $\theta = 180^\circ$

$$\lambda_f - \lambda_i = \Delta\lambda = \frac{h}{m_0 c} (1 - \cos 180^\circ)$$

$$\lambda_f - \lambda_i = \frac{h}{m_0 c} (1 - (-1)) \text{ since } \cos 180^\circ = -1$$

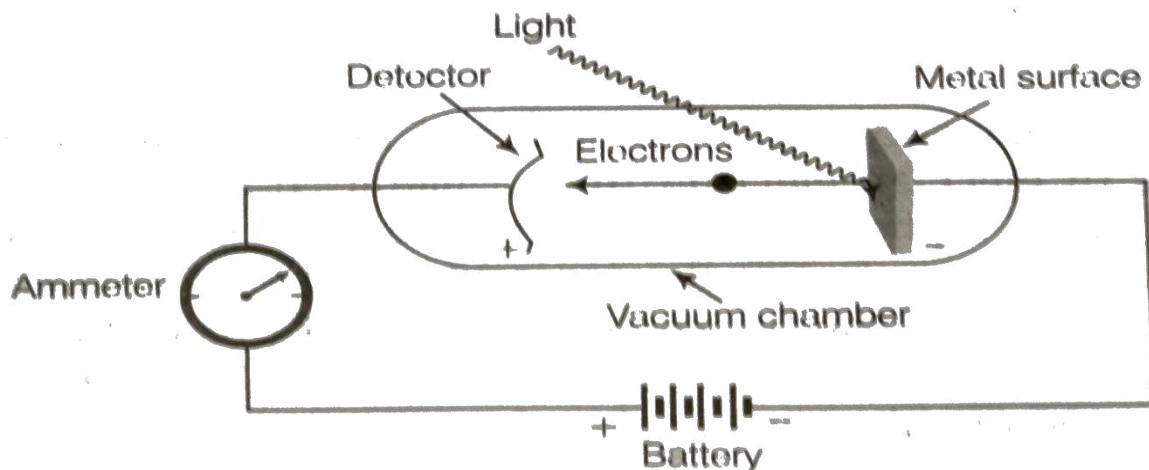
$$\lambda_f - \lambda_i = \frac{h}{m_0 c} (2)$$

$$\lambda_f - \lambda_i = \frac{2h}{m_0 c} = \frac{2 \times 6.625 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} = 0.4849 \text{Å}$$

Hence as θ varies from 0° to 180° , the wavelength of the scattered radiation increases from λ to $\lambda + \frac{2h}{m_0 c}$

1.5 Photoelectric effect:

The photoelectric effect is the emission of electrons when electromagnetic radiation, such as light, hits a material. Electrons emitted in this manner are called photoelectrons.



The apparatus used to study photo electric effect is shown above.

It consists of an evacuated tube which contains two electrodes. One is connected to the negative terminal of the battery while other is connected to the positive terminal of the battery through ammeter.

In the absence of light there is no flow of current .

When monochromatic light is allowed to fall on the metal surface the electrons are ejected and travel towards the detector. This results a current flow in the circuit.

The no.of ejected electrons and their kinetic energy depends upon the following factors :

- 1) The potential difference between the two plates.
- 2) The intensity of incident radiation.
- 3) The frequency of incident radiation.
- 4) The photo metal used.

Laws of photoelectric effect:

- (i) There is a definite cut off value of frequency below which electrons cannot be ejected by any substance.
- (ii) Number of emitted electrons are directly proportional to the intensity of light incident.
- (iii) Kinetic energy of emitted electrons depends on the frequency of incident light on substance.
- (iv) There is no time logging between the incident of light and emission of electrons.

1.6 Waves and Particles:

De Broglie suggested that the radiation has dual nature i.e both particle as well as wave nature. The concept of particle is easy to grasp. It has mass, velocity, momentum and energy. The concept of wave is a bit more difficult than that of a particle. A wave is spread out over a relatively large region of space, it cannot be said to be located just here and there, and it is hard to think of mass being associated with a wave. A wave is specified by its frequency, wavelength, phase, amplitude, intensity.

Considering the above facts, it appears difficult to accept the conflicting ideas that radiation has dual nature. However this acceptance is essential because radiation sometimes behaves as a wave and at other times as a particle.

- (1) Radiations behaves as waves in experiments based on interference, diffraction, polarization etc. this is due to the fact that these phenomena require the presence of two waves at the same position and at the same time. Thus we conclude that radiation behaves like wave.
- (2) Plank's quantum theory was successful in explaining blackbody radiation, photoelectric effect, Compton Effect and had established that the radiant energy, in its interaction with the matter, behaves as though it consists of corpuscles. Here radiation interacts with matter in the form of photons or quanta. Thus radiation behaves like particle.

Hence radiation cannot exhibit both particle and wave nature simultaneously.

1.7 de-Broglie hypothesis:

As electromagnetic waves behave like particles and Particles like electrons will behave like waves called **matter waves**. The matter waves thus conceived by de-Broglie are de-Broglie matter waves. He derived an expression for the wavelength of matter waves on the analogy of radiation

Based on the Planck's theory of radiation, the energy of a photon (Quanta) is given by

$$E = h\nu \quad \text{since } \nu = \frac{c}{\lambda}$$

$$E = h \frac{c}{\lambda} \dots \dots \dots (1)$$

$c \rightarrow$ is velocity of light

λ is wavelength

According to Einstein mass-energy relation

From eqns (1) and (2)

$$\frac{hc}{\lambda} = mc^2$$

$$\lambda = \frac{h}{mc} = \frac{h}{P}$$

Where: $P = mc \rightarrow$ is momentum of photon

$\lambda \rightarrow$ s de-Broglie wavelength associated with a photon.

De-Broglie proposed the concept of matter waves, according to which a material particle of mass 'm' moving with velocity ' v ' should be associated with de-Broglie wavelength ' λ ' given by

Wavelength for particle can be written as

$$\lambda = \frac{h}{mv} = \frac{h}{P}$$

Where: $P = mv \rightarrow$ momentum associated with particle

De-Broglie suggested that this equation for wavelength is a perfectly general one, applying to, material particles as well as to photons.

If E is the kinetic energy of particle

$$E = \frac{1}{2}mv^2 = \frac{1}{2} \frac{m^2 v^2}{m} = \frac{1}{2} \frac{P^2}{m}$$

$$P = \sqrt{2mE}$$

Applied Physics

I -B.Tech

Hence de-Broglie's wave length is

$$\lambda = \frac{h}{\sqrt{2mE}}$$

de-Broglie's wave length of Electron

If m_0 is the rest mass of electron and it is accelerated with a potential V . if v is the velocity attained by electron due to acceleration.

Kinetic energy of electron

$$E = \frac{1}{2} m_0 v^2 \dots \dots \dots 1$$

Energy of electron due to application of potential

$$E = eV \dots \dots \dots 2$$

From 1 and 2

$$\frac{1}{2} m_0 v^2 = eV$$

$$\frac{1}{2} m_0^2 v^2 = m_0 eV$$

$$m_0^2 v^2 = 2m_0 eV$$

$$P^2 = 2m_0 eV$$

From de-Broglie wave length

$$\lambda = \frac{h}{mc} = \frac{h}{P}$$

$$\lambda = \frac{h}{\sqrt{2m_0 eV}}$$

$$\lambda = \frac{h}{\sqrt{2 * V * 9.1 * 10^{-31} * 1.63 * 10^{-19}}}$$

$$\lambda = \frac{12.26}{\sqrt{V}} \text{ \AA}$$

The above expression is for non relativistic case since relative variation of mass with velocity is not considered. Thus accelerated electrons exhibit wave nature corresponding to wave length of x rays. This concept only helped Davisson & Germer to provide experimental evidence on matter waves when they conducted electron diffraction experiments.

Characteristics of Matter waves:

Since $\lambda = h / m v$

1. Lighter the particle, greater is the wavelength associated with it.
2. Lesser the velocity of the particle, longer the wavelength associated with it.
3. For $v = 0$, $\lambda = \infty$. This means that only with moving particle, matter waves is associated.
4. Whether the particle is changed or not, matter waves is associated with it.
5. It can be proved that matter waves travel faster than light.

We know that $E = h \nu$ and $E = m c^2$

$$h\nu = mc^2 \text{ or } \nu = mc^2 / h$$

Wave velocity (ω) is given by

$$\omega = \nu \lambda = mc^2 \lambda / h = (mc^2 / h) (h / m \nu)$$

$$\Rightarrow \omega = c^2 / \nu$$

As the particle velocity 'v' cannot exceed velocity of light, ω is greater than the velocity of light.

6. No single phenomena exhibit both particle nature and wave nature simultaneously.
7. The wave nature of matter introduces an uncertainty in the location of the particle & the momentum of the particle exists when both are determined simultaneously.

1.8 Davisson and Germer's experiment:

C. J. Davisson and L. H. Germer were studying scattering of electrons by a metal target and measuring the intensity of electrons scattered in different directions.

Experimental Arrangement:

An electron gun, which comprises of a tungsten filament is heated by a low tension battery B1, produces electrons. These electrons are accelerated to desired velocity by applying suitable potential from a high tension source B2. The accelerated electrons are collimated into a fine beam by allowing them to pass through a system of pin holes provided in the cylinder. The whole instrument is kept in an evacuated chamber.

The past moving beam of electrons is made to strike the Nickel target capable of rotating about an axis perpendicular to the plane. The electrons are now scattered in all directions by the atomic planes of crystals. The intensity of the electron beam scattered in a direction can be measured by the electron collector which can be rotated about the same axis as the target. The collector is connected to a galvanometer whose deflection is proportional to the intensity of the electron beam entering the collector.

The electron beam is accelerated by 54 V is made to strike the Nickel crystal and a sharp maximum is occurred at angle of 50° with the incident beam. The incident beam and the diffracted beam in this experiment make an angle of 65° with the family of Bragg's planes.

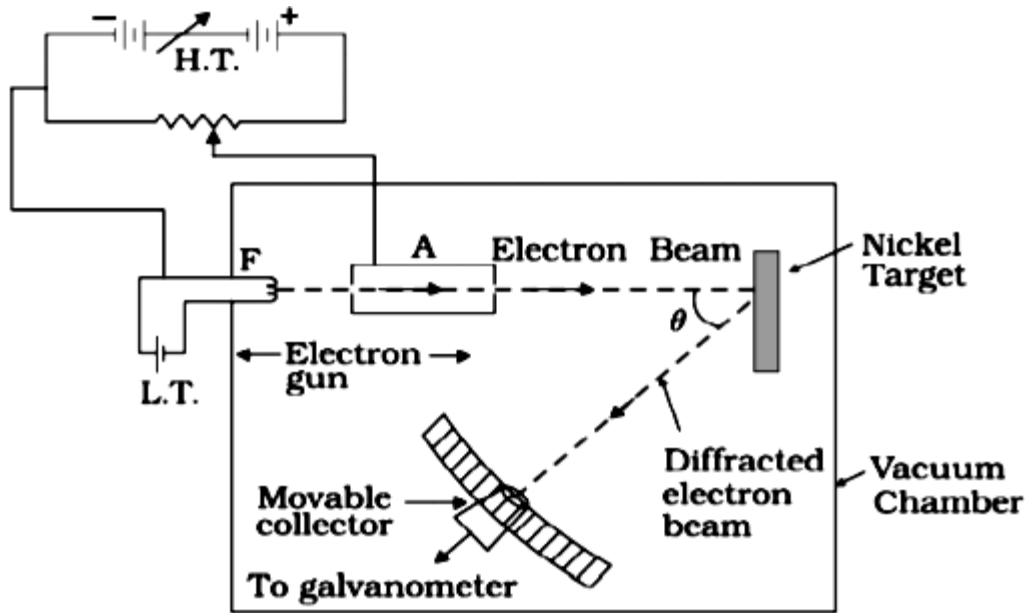


Fig. Davisson and Germer's Apparatus

$$d = 0.091 \text{ nm} \quad (\text{for Ni crystals})$$

According to Bragg's law for maxima in diffracted pattern,

$$2d \sin \theta = n \lambda$$

$$\text{For } n=1, \quad \lambda = 2d \sin \theta$$

$$= 2 \times 0.091 \times 10^{-10} \times \sin 65^\circ$$

$$= 0.165 \text{ nm}$$

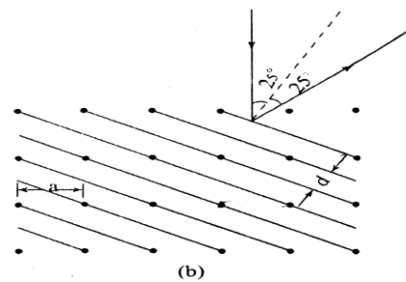
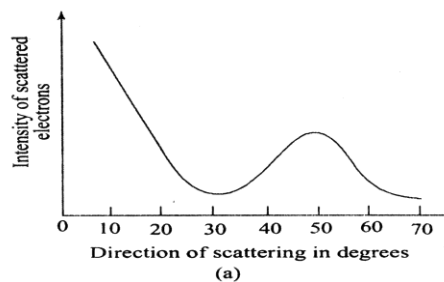


Figure : (a) The variation of the intensity of scattered electrons with the direction of scattering and (b) nickel crystal acting as a grating

For a 54 V electron, the deBroglie wavelength associated with the electron is given by

$$= 12.25 / \sqrt{V} = (12.25 / \sqrt{54})^\circ \text{A}$$

$$= 0.166 \text{ nm.}$$

This value is in agreement with the experimental value. This experiment provides a direct verification of deBroglie hypothesis of wave nature of moving particles.

1.9 Heisenberg Uncertainty Principle:

According to classical mechanics, a moving particle at any instant has a fixed position in space and a definite momentum which can be determined simultaneously with any desired accuracy. The classical point of view represents an approximation which is adequate for the objects of appreciable size, but not for the particles of atomic dimensions.

Since a moving particle has to be regarded as a de Broglie group, there is a limit to the accuracy with which we can measure the particle properties. The particle may be found anywhere within the wave group, moving with the group velocity. If the group is narrow, it is easy to locate its position but the uncertainty in calculating its velocity or momentum increases. On the other hand, if the group is wide, its momentum can be estimated satisfactorily, but the uncertainty in finding the location of the particle is great. Heisenberg stated that the simultaneous determination of exact position and momentum of a moving particle is impossible.

If Δx is Error in the measurement of position of the particle along X-axis

Δp is Error in the measurement of momentum

Then $\Delta x \cdot \Delta p = h$ ----- (1) where h is Plank's constant

The above relation represents the uncertainty involved in measurement of both the position and momentum of the particle.

To optimize the above error, lower limit is applied to the eqn. (1)

$$\text{Then } (\Delta x) \cdot (\Delta p) \geq \hbar / 2 \text{ where } \hbar = h / 2 \Pi$$

A particle can be exactly located ($\Delta x \rightarrow 0$) only at the expense of an infinite momentum ($\Delta p \rightarrow \infty$).

There are uncertainty relations between position and momentum, energy and time, and angular momentum and angle.

If the time during which a system occupies a certain state is not greater than Δt , then the energy of the state cannot be known within ΔE ,

$$\text{i.e. } (\Delta E) (\Delta t) \geq \hbar / 2 .$$

1.10 Schrodinger's Time Independent Wave Equation:

Schrödinger, in 1926, developed wave equation for the moving particles. One of its forms can be derived by simply incorporating the de-Broglie wavelength expression into the classical wave eqn.

If a particle of mass moving with velocity 'v' is associated with a group of waves.

Let ψ be the wave function of the particle. Also let us consider a simple form of progressing wave like the one represented by the following equation,

$$\psi = \psi_0 \sin(\omega t - kx) \dots \dots \dots (1)$$

Where $\psi = \psi(x, t)$ and ψ_0 is the amplitude.

Differentiating ψ partially w.r.to x ,

$$\begin{aligned} \frac{\partial \psi}{\partial x} &= \psi_0 \cos(\omega t - kx)(-k) \\ &= -k\psi_0 \cos(\omega t - kx) \end{aligned}$$

Once again differentiate w.r.to x

$$\begin{aligned} \frac{\partial^2 \psi}{\partial x^2} &= -k\psi_0 (-\sin(\omega t - kx))(-k) \\ &= -k^2\psi_0 \sin(\omega t - kx) \\ \frac{\partial^2 \psi}{\partial x^2} &= -k^2\psi \quad \text{from eq (1)} \\ \frac{\partial^2 \psi}{\partial x^2} + k^2\psi &= 0 \dots \dots \dots (2) \end{aligned}$$

Since $k = \frac{2\pi}{\lambda}$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{4\pi^2}{\lambda^2} \psi = 0 \dots \dots \dots (3)$$

From eqn. (2) or eqn. (3) is the differential form of the classical wave eqn. now we incorporate de-Broglie wavelength expression $\lambda = \frac{h}{mv}$

Thus we obtain

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{4\pi^2}{\left(\frac{h}{mv}\right)^2} \psi = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{4\pi^2 m^2 v^2}{h^2} \psi = 0 \dots \dots \dots (4)$$

The total energy E of the particle is the sum of its kinetic energy K and potential energy V

$$E = K.E. + P.E \dots \dots \dots (5)$$

$$E = \frac{1}{2} m v^2 + V$$

$$E - V = \frac{1}{2} m v^2 \dots \dots \dots (6)$$

$$2(E - V) = m v^2$$

$$2m(E - V) = m^2 v^2 \dots \dots \dots (7)$$

From (4) and (7)

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{8\pi^2 m(E - V)}{h^2} \psi = 0 \dots \dots \dots (8)$$

In quantum mechanics, the value $\frac{h}{2\pi}$ occurs more frequently.

Hence we denote, $\hbar = \frac{h}{2\pi}$

Using this notation, we have

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m(E - V)}{\hbar^2} \psi = 0 \dots \dots \dots (9)$$

For simplicity, we considered only one – dimensional wave. Extending eqn. (9) for a three – dimensional, we have

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m(E - V)}{\hbar^2} \psi = 0 \dots \dots \dots (10)$$

Where $\psi = \psi(x, y, z)$

Here, we have considered only stationary states of ψ after separating the time dependence of ψ .

Using the Laplacian operator,

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \dots \dots \dots (11)$$

Eqn. (10) can be written as

$$\nabla^2 \psi + \frac{2m(E - V)}{\hbar^2} \psi = 0 \dots \dots \dots (12)$$

This is the Schrödinger Time Independent Wave Equation.

Physical Significance of Wave Function:

Max Born in 1926 gave a satisfactory interpretation of the wave function ψ associated with a moving particle. He postulated that the square of the magnitude of the wave function $|\psi|^2$ (or $\psi\psi^*$), evaluated at a particular point represents the probability of finding the particle at the point. $|\psi|^2$ is called the probability density and ψ is the probability amplitude. Thus the probability of the particle within an element volume $d\tau$ is $|\psi|^2 d\tau$. Since the particle is certainly somewhere, the integral at $|\psi|^2 d\tau$ over all space must be unity i.e.

$$\int_{-\infty}^{\infty} |\psi|^2 .d\tau = 1$$

A wave function that obeys the above equations is said to be normalized. Energy acceptable wave function must be normalizable besides being normalizable; an acceptable wave function should fulfill the following requirements (limitations)

It must be finite everywhere.

It must be single valued.

It must be continuous and have a continuous first derivative everywhere.

Normalization of a wave function:

Since $|\psi(x, y, z)|^2 .dv$ is the probability that the particle will be found in a volume element dv . Surrounding the point at position (x, y, z) , the total probability that the particle will be somewhere in space must be equal to 1. Thus, we have

$$\int_{-\infty}^{\infty} |\psi(x, y, z)|^2 .dv = 1$$

Where ψ is a function of the space coordinates (x, y, z) from this 'normalization condition' we can find the value of the constant and its sign. A wave function which satisfies the above condition is said to be normalized (to unity).

The normalizing condition for the wave function for the motion of a particle in one dimension is

$$\int_{-\infty}^{\infty} |\psi(x)|^2 .dx = 1$$

From these equations, we see that for one – dimensional case, the dimension of $\psi(x)$ is $L^{-1/2}$ and for the three dimensional case the dimension of $\psi(x, y, z)$ is $L^{-3/2}$.

1.11 Particle in One Dimensional Potential Box:

Consider a particle of mass 'm' placed inside a one-dimensional box of infinite height and width L.

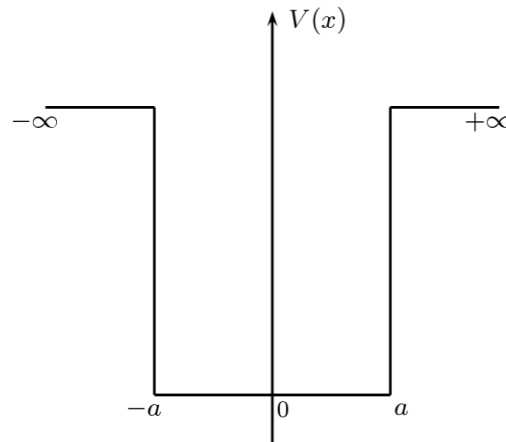


Fig. Particle in a potential well of infinite height.

Assume that the particle is freely moving inside the box. The motion of the particle is restricted by the walls of the box. The particle is bouncing back and forth between the walls of the box at $x = 0$ and $x = a$. For a freely moving particle at the bottom of the potential well, the potential energy is very low. Since the potential energy is very low, moving particle energy is assumed to be zero between $x = 0$ and $x = a$.

The potential energy of the particle outside the walls is infinite due to the infinite P.E.

The particle cannot escape from the box

$$\text{i.e. } V = 0 \quad \text{for } 0 < x < a$$

$$V = \infty \quad \text{for } x \leq 0 \text{ or } x \geq a$$

Since the particle cannot be present outside the box, its wave function is zero

$$\text{i.e. } |\psi| = 0 \quad \text{for } x < 0 \text{ or } x > a$$

$$|\psi| = 0 \quad \text{for } x = 0 \text{ and } x = a$$

The Schrödinger one – dimensional time independent eqn. is

$$\frac{\partial^2 \psi}{\partial x^2} + [2m(E - V) / \hbar^2] \psi = 0 \quad \text{-----(1)}$$

For freely moving particle $V = 0$

$$\frac{\partial^2 \psi}{\partial x^2} + [2mE / \hbar^2] \psi = 0 \quad \text{-----(2)}$$

Applied Physics

I -B.Tech

$$\text{Taking } 2mE/\hbar^2 = K^2 \text{ ----- (3)}$$

$$\text{Eqn.(1) becomes } \partial^2 \Psi / \partial x^2 + k^2 \Psi = 0 \text{ -----(4)}$$

Eqn. (1) is similar to eq. of harmonic motion and the solution of above eqn. is written as

$$\Psi = A \sin kx + B \cos kx \text{ -----(5)}$$

where A, B and k are unknown quantities and to calculate them it is necessary to construct boundary conditions.

Hence boundary conditions are

$$\text{When } x = 0, \Psi = 0 \Rightarrow \text{from (5)} 0 = 0 + B \Rightarrow B = 0 \text{ ----- (6)}$$

$$\text{When } x = a, \Psi = 0 \Rightarrow \text{from (5)} 0 = A \sin ka + B \cos ka \text{ ----- (7)}$$

But from (6) $B = 0$ therefore eqn. (7) may turn as

$$A \sin ka = 0$$

Since the electron is present in the box $a \neq 0$

$$\sin ka = 0$$

$$Ka = n\pi$$

$$k = n\pi / a \text{ ----- (8)}$$

Substituting the value of k in eqn. (3)

$$2mE/\hbar^2 = (n\pi/a)^2$$

$$E = (n\pi/a)^2 (\hbar^2/2m) = (n\pi/a)^2 (h^2/8m\pi^2)$$

$$E = n^2 h^2 / 8ma^2$$

$$E = n^2 h^2 / 8ma^2$$

$$\text{In general } E_n = n^2 h^2 / 8ma^2 \text{ -----(9)}$$

The wave eqn. can be written as

$$\Psi = A \sin (n\pi x / a) \text{ ----- (10)}$$

Let us find the value of A, if an electron is definitely present inside the box, then

$$\Rightarrow \int_{-\infty}^{\infty} |\Psi|^2 dx = 1$$

$$\Rightarrow \int_0^a A^2 \sin^2 (n\pi x / a) dx = 1$$

$$\Rightarrow \int_0^a \sin^2 (n\pi x / a) dx = 1 / A^2$$

$$\Rightarrow \int_0^a [1 - \cos (2\pi n (x/a)) / 2] dx = 1 / A^2$$

Applied Physics

I -B.Tech

$$A = \sqrt{2/a} \quad \text{----- (11)}$$

From eqn's. (10) & (11)

$$\Psi_n = \sqrt{2/a} \sin(n\pi x/a) \quad \text{----- (12)}$$

Eqn. (9) represents an energy level for each value of n. The wave function for this energy level is given in eqn. (12). Therefore the particle in the box can have discrete values of energies. These values are quantized. Not that the particle cannot have zero energy. The normalized wave functions Ψ_1, Ψ_2, Ψ_3 given by eqn (12) is plotted. The values corresponding to each E_n value is known as Eigen value and the corresponding wave function is known as Eigen function.

The wave function Ψ_1 , has two nodes at $x = 0$ & $x = a$

The wave function Ψ_2 , has three nodes at $x = 0, x = a/2$ & $x = a$

The wave function Ψ_3 , has three nodes at $x = 0, x = a/3, x = 2a/3$ & at $x = a$

The wave function Ψ_n , has $(n + 1)$ nodes

Substituting the value of E in (3), we get

For three dimensional potential box, the eigen functions are given by

$$\Psi_{n2} \sin k_1 x \times \sin k_2 y \times \sin k_3 z$$

$$\text{Where } k_1 = \frac{n_1\pi}{a}, k_2 = \frac{n_2\pi}{a}, k_3 = \frac{n_3\pi}{a},$$

$$\text{Eigen values are given by } E_n = \frac{n^2 h^2 \pi^2}{8ma^2}$$

Eigen values are given by

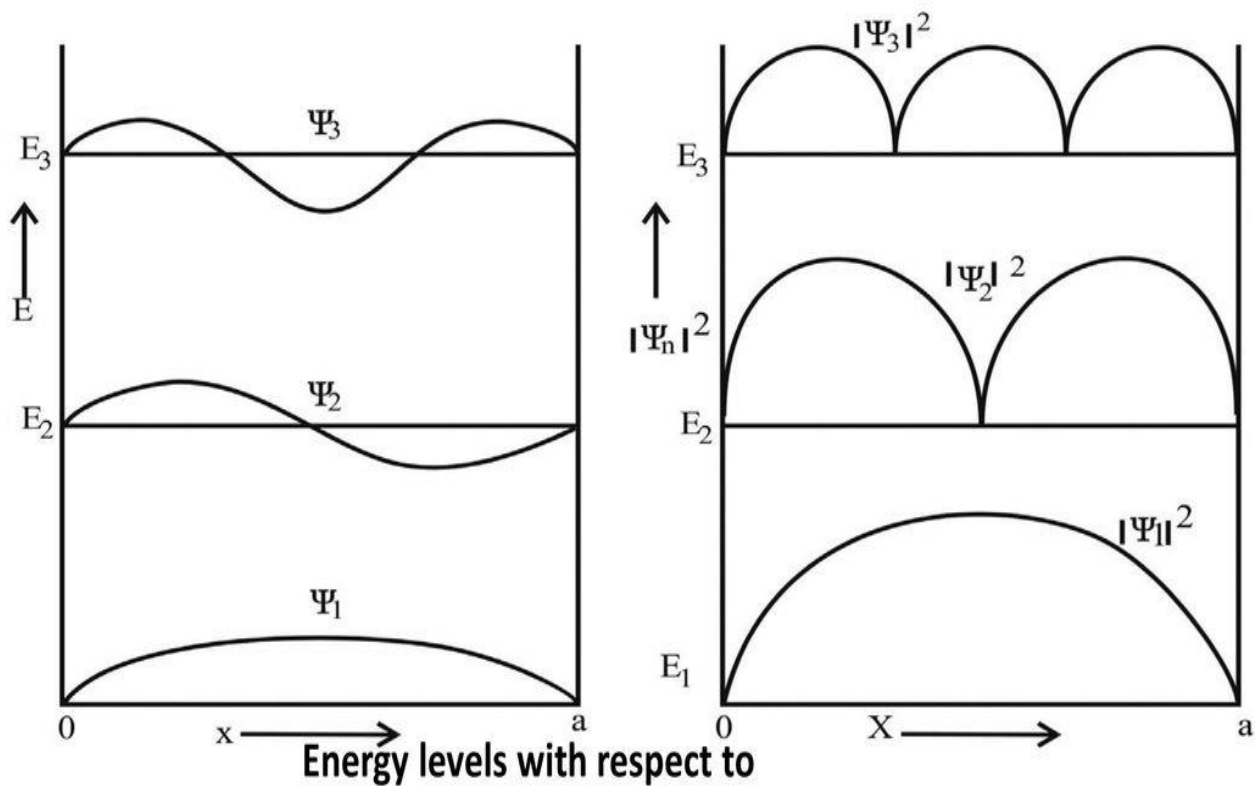
$$(2m/\hbar^2)(p^2/2m) = k^2$$

$$\text{Where } n^2 = n_1^2 + n_2^2 + n_3^2$$

$$\Rightarrow p^2/\hbar^2 = k^2$$

$$k = p/\hbar = p/(h/2\pi) = 2\pi p/h$$

$$k = 2\pi/\lambda \quad \text{where } k \text{ is known as wave vector.}$$



(a) wave functions and (b) probability density

Therefore, the normalized wave function is given as,

$$\Psi_n = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x$$

Applied Physics

I -B.Tech

1.	Equation for wavelength of a Photon is $\lambda = \frac{h}{mc}$	6.	Eigen energy level of a particle is given by $E_n = \frac{n^2 h^2}{8mL^2}$
2.	De Broglie's wavelength for a particle associated with some velocity is $\lambda = \frac{h}{mv}$	7.	Equation for the particle in a one dimensional box is given by $\psi = (\frac{\sqrt{2}}{L}) \sin \frac{n\pi x}{L}$
3.	De Broglie's wavelength for a particle with some energy is $\lambda = \frac{h}{\sqrt{2mE}}$	8.	Heisenberg's Uncertainty Principle for position & momentum is given by $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$
4.	De Broglie's wavelength for a particle associated with T is $\lambda = \frac{h}{\sqrt{3kT}}$	9.	Schrodinger wave equation(time independent) is $\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0$
5.	De Broglie's wavelength associated with an electron with potential is $\lambda = \frac{1.226}{\sqrt{V}} \text{nm}$	10.	Compton shift $\lambda_f - \lambda_i = \Delta\lambda = \frac{h}{m_0 c} (1 - \cos \theta)$

ELECTRON THEORY OF SOLIDS

1. 12 FREE ELECTRON THEORY

In solids, electrons in outer most orbits of atoms determine its electrical properties. Electron theory is applicable to all solids, both metals and non metals. In addition, it explains the electrical, thermal and magnetic properties of solids. The structure and properties of solids are explained employing their electronic structure by the electron theory of solids.

It has been developed in three main stages:

1. Classical free electron theory
2. Quantum Free Electron Theory.
3. Zone Theory.

Classical free electron theory: The first theory was developed by Drude & Lorentz in 1900. According to this theory, metal contains free electrons which are responsible for the electrical conductivity and metals obey the laws of classical mechanics.

Quantum Free Electron Theory: In 1928 Sommerfield developed the quantum free electron theory. According to Sommerfield, the free electrons move with a constant potential. This theory obeys quantum laws.

Zone Theory: Bloch introduced the band theory in 1928. According to this theory, free electrons move in a periodic potential provided by the lattice. This theory is also called “Band Theory of Solids”. It gives complete informational study of electrons.

Classical free electron theory:

Even though the classical free electron theory is the first theory developed to explain the electrical conduction of metals, it has many practical applications. The advantages and disadvantages of the classical free electron theory are as follows:

Advantages:

1. It explains the electrical conductivity and thermal conductivity of metals.
2. It verifies ohm's law.
3. It is used to explain the optical properties of metals.
4. Metal composed of atoms in which electrons revolve around the nucleus are many states available for occupation. If the density of states is zero, no states can be occupied at that energy level.
5. The valence electrons are freely moving about the whole volume of the metals like the molecules of perfect gas in a container

Applied Physics

I -B.Tech

6. The free electrons moves in random directions and collide with either positive ions or other free electrons. Collision is independent of charges and is elastic in nature
7. The movements of free electrons obey the laws of classical kinetic theory of gases
8. Potential field remains constant throughout the lattice.
9. In metals, there are large numbers of free electrons moving freely within the metal i.e. the free electrons or valence electrons are free to move in the metal like gaseous molecules, because nuclei occupy only 15% metal space and the remaining 85% space is available for the electrons to move.

Drawbacks:

1. It fails to explain the electric specific heat and the specific heat capacity of metals.
2. It fails to explain superconducting properties of metals.
3. It fails to explain new phenomena like photoelectric effect, Compton effect, black – Body radiation, etc.
4. It fails to explain Electrical conductivity (perfectly) of semiconductors or insulators.
5. The classical free electron model predicts the incorrect temperature dependence of
6. It fails to give a correct mathematical expression for thermal conductivity.
7. Ferromagnetism couldn't be explained by this theory.
8. Susceptibility has greater theoretical value than the experimental value.

Quantum free electron theory:

Advantages:

1. All the electrons are not present in the ground state at 0 K, but the distribution obeys Pauli's exclusion principle. At 0 K, the highest energy level filled is called Fermi- level.
2. The potential remains constant throughout the lattice.
3. Collision of electrons with positive ion cores or other free electrons is independent of charges and is elastic in nature. Energy levels are discrete.
4. It was successful to explain not only conductivity, but also thermionic emission paramagnetism, specific heat.

Drawbacks:

1. It fails to explain classification of solids as conductors, semiconductors and insulators.

1.14 STATISTICAL DISTRIBUTION FUNCTIONS:

The main assumption of Maxwell –Boltzmann statistics are

1. The particles are identical & distinguishable.

Applied Physics

I -B.Tech

2. The volume of each phase space cell chosen is extremely small and hence chosen volume has very large no of cells.
3. Since cells are extremely small, each cell can have either one particle or no particle though there is no limit on the number of particles which can occupy a phase space cell.
4. System is isolated which means both total Energy and total no of particles remain constant.
5. State of each particle is specified either by its cell number in phase. Space or instantaneous position and momentum coordinate.
6. Energy levels are continuous.
7. Maxwell distribution function $F_{MB} = A / \exp(-E_i/kt)$

Where E_i – i th Energy level.

k - Boltzmann constant

T - Absolute temp

8. The particles which obey M-B statistics system consisting of molecules.

Bose - Einstein statistics:

The main assumption of B-E statistics:

- 1) The particles which obey B-E statistics are called bosons.
- 2) The Bosons of system are identical & indistinguishable
- 3) Bosons obey uncertainty principle.
- 4) Any number of bosons can occupy a single cell in phase space.
- 5) Bosons do not obey Pauli's Exclusion principle
- 6) The numbers of phase space cells is comparable with the numbers of bosons .
- 7) wave functions representing the Bosons are symmetric ie $\psi(2,1) = \psi(1,2)$
- 8) The wave function s of boson do overlap slightly ie weak interaction exists.
- 9) Energy states are discrete

$$i. f_{BE}(E) = \frac{1}{\exp\left(\frac{E_F - E_i}{kt}\right) - 1}$$

E_i – i^{th} energy level

E_F - Fermi Energy

T - Absolute temp and k- Boltzmann const.

The particles which obey boson are photons, He_4 , π and K- mesons, phonon in a solids, Bosons can have spin of $nh/2\pi$ where n is integer.

Fermi – Dirac Statistics:

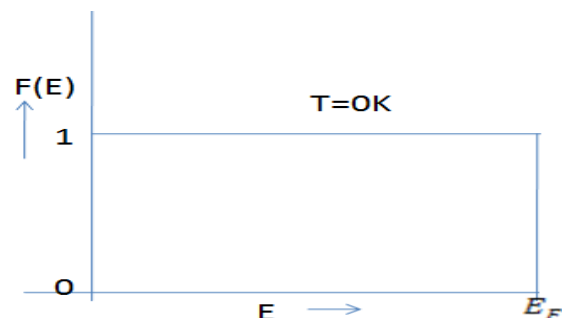
- 1) The particles which obey F –D statistics are called fermions.
- 2) Fermions are identical and indistinguishable
- 3) There cannot be more than one particle in a single cell in phase space i.e they obey Pauli's exclusion principle.
- 4) Fermions have integral spin
- 5) Wave function representing fermions are antisymmetric ie $\varphi(1,2) = - \varphi(2,1)$
- 6) Weak interaction exists between particles
- 7) Uncertainty principle is applicable.
- 8) Energy states are discrete.
- 9) The Fermi Dirac distribution function $f(D)$, is the probability that a fermions occupies a state of energy E and is given by

$$f_{FD}(\varepsilon) = \frac{1}{\exp\left(\alpha + \frac{E}{kt}\right) + 1}$$

α is constant and is dependent on temperature $\alpha = -E_F/kT$

1.15 Fermi level and Fermi energy

The distribution of energy states in a semiconductor is explained by Fermi-Dirac statistics since it deals with the particles having half integral spin like electrons. Consider that the assembly of electrons as electron gas which behaves like a system of Fermi particles or fermions. The Fermions obeying Fermi–Dirac statistics i.e., Pauli's exclusion principle.



Applied Physics

I -B.Tech

Fermi energy: It is the energy of state at which the

Probability of electron occupation is $\frac{1}{2}$ at any temperature above 0K. It separates filled energy states and unfilled energy states. The highest energy level that can be occupied by an electron at 0 K is called Fermi energy level

Fermi level: It is a level at which the electron probability is $\frac{1}{2}$ at any temp above 0K (or) always it is 1 or 0 at 0K.

Therefore, the probability function $F(E)$ of an electron occupying an energy level E is given by,

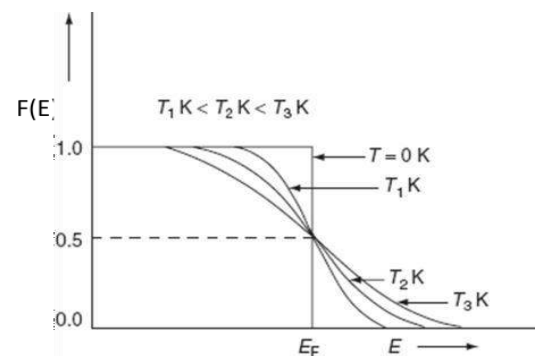
$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)} \quad \text{----- (1)}$$

Where E_F known as Fermi energy and it is constant for a system,

k is the Boltzmann constant and T is the absolute temperature.

Case I : Probability of occupation at $T = 0K$, and $E < E_F$ Therefore $F(E) = 1$, as per above, clearly indicates that at $T = 0K$, the energy level below the Fermi energy level E_F is fully occupied by electrons leaving the upper level vacant.

Therefore, there is 100% probability that the electrons to occupy energy level below Fermi level.



Case II: Probability of occupation at $T = 0K$, and $E > E_F$ i.e., all levels below E_F are completely filled and all levels above E_F are completely empty. As the temperature rises $F(E)$.

Case III: Probability of occupation at $T = 0K$, and $E = E_F$

$$F(E) = \frac{1}{1 + \exp^0} = \frac{1}{1 + 1} = 0.5$$

The above condition states that, $T = 0K$, there is a 50% probability for the electrons to occupy Fermi energy.

The probability function $F(E)$ lies between 0 and 1

Hence there are three probabilities namely

1. $F(E) = 1$ 100% probability to occupy the energy level by electrons.
2. $F(E) = 0$ No probability to occupy the energy levels by electrons and hence ,it is empty.
3. $F(E) = 0.5$ 50% probability of finding the electron in the energy level.

1.16 PERIODIC POTENTIAL AND BLOCH THEOREM:

According to free electron model, a conduction electron in metal experiences constant potential. But in real crystal, there exists a periodic arrangement of positively charged ions through which the electrons move. As a consequence, the potential experienced by electrons is not constant but it varies with the periodicity of the lattice. In zone theory, as per Bloch, potential energy of electrons considered as varying potential with respect to lattice 'a'.

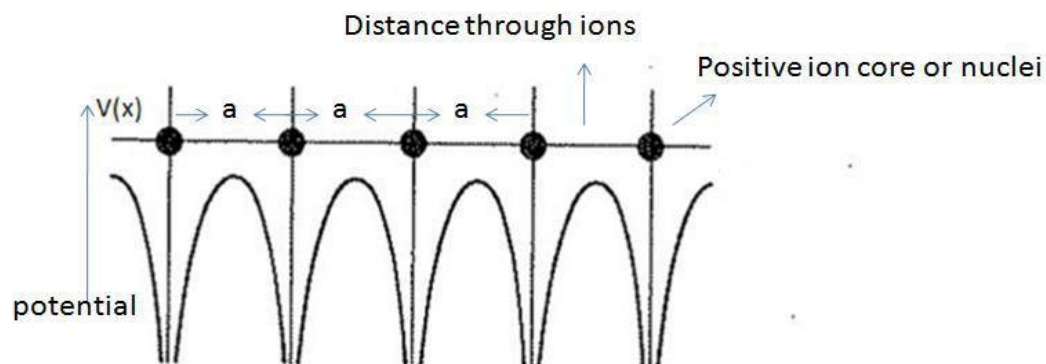
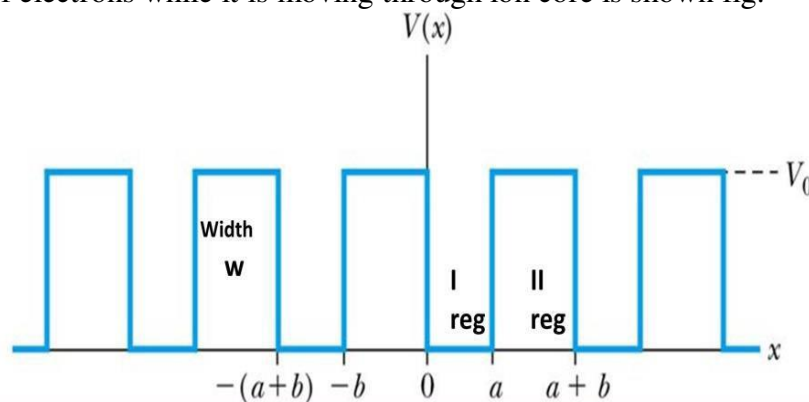


Fig: Variation of potential energy in a periodic lattice

1.17 KRONIG -PENNEY MODEL:

According to this theory, the electrons move in a periodic potential produced by the positive ion cores. The potential of electron varies periodically with periodicity of ion core and potential energy of the electrons is zero near nucleus of the positive ion core. It is maximum when it is lying between the adjacent nuclei which are separated by inter-atomic spacing. The variation of potential of electrons while it is moving through ion core is shown fig.



Applied Physics

I -B.Tech

$$V(x) = \{ 0, \text{ for the region } 0 < x < a$$

$$\{ V_0 \text{ for the region } -b < x < a \text{ -----(1)}$$

Applying the time independent Schrödinger's wave equation for above two regions

$$d^2\Psi / dx^2 + 2 m E \Psi / \hbar^2 = 0 \text{ for region } 0 < x < a \text{ -----(2)}$$

$$\text{and } d^2\Psi / dx^2 + 2 m (E - V) \Psi / \hbar^2 = 0 \text{ for region } -b < x < a \text{ -----(3)}$$

$$\text{Substituting } \alpha^2 = 2 m E / \hbar^2 \text{ -----(4)}$$

$$\beta^2 = 2 m (E - V) / \hbar^2 \text{ -----(5)}$$

$$d^2\Psi / dx^2 + \alpha^2 \Psi = 0 \text{ for region } 0 < x < a \text{ -----(6)}$$

$$d^2\Psi / dx^2 + \beta^2 \Psi = 0 \text{ for region } -b < x < a \text{ -----(7)}$$

The solution for the eqn.s (6) and (7) can be written as

$$\Psi(x) = U_k(x) e^{ikx} \text{ -----(8)}$$

The above solution consists of a plane wave e^{ikx} modulated by the periodic function.

$U_k(x)$, where this $U_k(x)$ has the periodicity of the ion such that

$$U_k(x) = U_k(x+a) \text{ -----(9)}$$

and where k is propagating vector along x -direction and is given by $k = 2 \Pi / \lambda$. This k is also known as wave vector.

Differentiating equation (8) twice with respect to x , and substituting in equation (6) and (7), two independent second order linear differential equations can be obtained for the regions $0 < x < a$ and $-b < x < 0$.

Applying the boundary conditions to the solution of above equations, for linear equations in terms of A, B, C and D it can be obtained (where A, B, C, D are constants) the solution for these equations can be determined only if the determinant of the coefficients of A, B, C , and D vanishes, on solving the determinant.

$$(\beta^2 - \alpha^2 / 2 \alpha \beta) \sin \hbar \beta b \sin \alpha a + \cos \hbar \beta b \cos \alpha a = \cos k(a + b) \text{ ----- (10)}$$

The above equation is complicated and Kronig and Penney could conclude with the equation. Hence they tried to modify this equation as follows

Let V_0 is tending to infinite and b is approaching to zero. Such that $V_0 b$ remains finite. Therefore $\sin \hbar \beta b \rightarrow \beta b$ and $\cos \hbar \beta b \rightarrow 1$

$$\begin{aligned} \beta^2 - \alpha^2 &= (2 m / \hbar^2) (V_0 - E) - (2 m E / \hbar^2) \\ &= (2 m / \hbar^2) (V_0 - E - E) = (2 m / \hbar^2) (V_0 - 2 E) \end{aligned}$$

$$= 2 m V_o / \hbar^2 \text{ (since } V_o \gg E \text{)}$$

Substituting all these values in equation (10) it verifies as

$$(2 m V_o / 2 \hbar^2 \alpha \beta) \beta b \cdot \sin \alpha a + \cos \alpha a = \cos k a$$

$$(m V_o b a / \hbar^2) (\sin \alpha a / \alpha a) + \cos \alpha a = \cos k a$$

$$(P / \alpha a) \sin \alpha a + \cos \alpha a = \cos k a \text{ -----(11)}$$

Where $P = [m V_o b a / \hbar^2] \text{ -----(12)}$

and is a measure of potential barrier strength.

The left hand side of the equation (11) is plotted as a function of α for the value of $P = 3 \Pi / 2$ which is shown in fig, the right hand side one takes values between -1 to +1 as indicated by the horizontal lines in fig. Therefore the equation (11) is satisfied only for those values of ka for which left hand side between ± 1 .

From fig , the following conclusions are drawn.

- 1) The energy spectrum of the electron consists of a number of allowed and forbidden energy bands.
- 2) The width of the allowed energy band increases with increase of energy values ie increasing the values of αa . This is because the first term of equation(11) decreases with increase of αa .

$$(P / \alpha a) \sin \alpha a + \cos \alpha a = 3 \Pi / 2$$

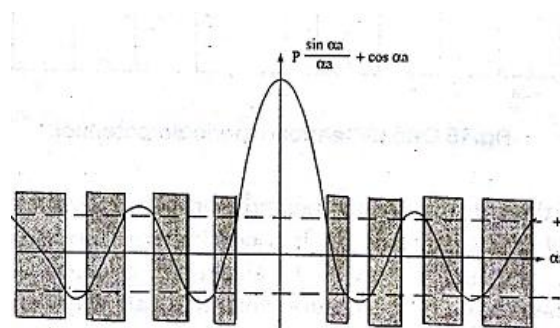


Fig.16 Plot of $P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a$ versus αa .

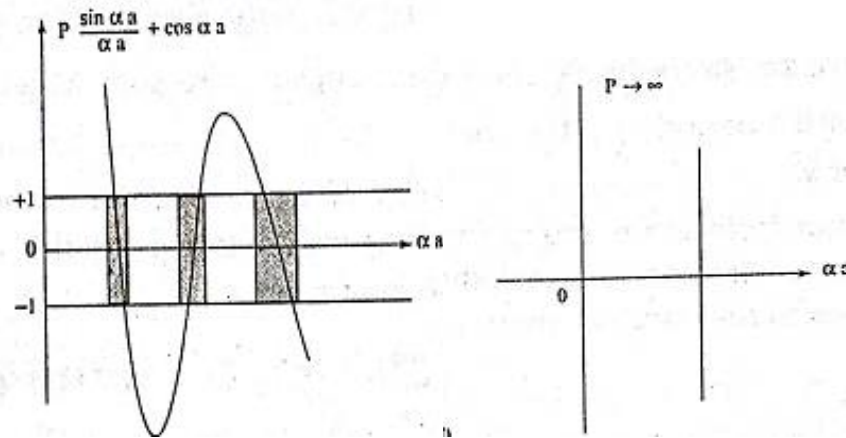


Fig. $p \rightarrow \infty$ and $p \rightarrow 0$

- 3) With increasing P , ie with increasing potential barrier, the width of an allowed band decreases. As $P \rightarrow \infty$, the allowed energy becomes infinitely narrow and the energy spectrum is a line spectrum as shown in fig.

If $P \rightarrow \infty$, then the equation (11) has solution ie $\sin \alpha a = 0$

$$\alpha a = \pm n \Pi$$

$$\alpha = \pm n \Pi / a$$

$$\alpha^2 = n^2 \Pi^2 / a^2$$

$$\text{But } \alpha^2 = 2 m E / \hbar^2$$

$$\text{Therefore } 2 m E / \hbar^2 = n^2 \Pi^2 / a^2$$

$$E = [\hbar^2 \Pi^2 / 2 m a^2] n^2$$

$$E = n \hbar^2 / 8 m a^2 \quad (\text{since } \hbar = h / 2 \Pi)$$

This expression shows that the energy spectrum of the electron contains discrete energy levels separated by forbidden regions.

- 4) When $P \rightarrow 0$ then $\cos \alpha a = \cos ka$

$$\alpha = k, \quad \alpha^2 = k^2$$

$$\text{but } \alpha^2 = 2 m E / \hbar^2$$

$$k^2 = (h^2 / 2 m) (1 / \lambda^2) = (h^2 / 2 m) (P^2 / h^2)$$

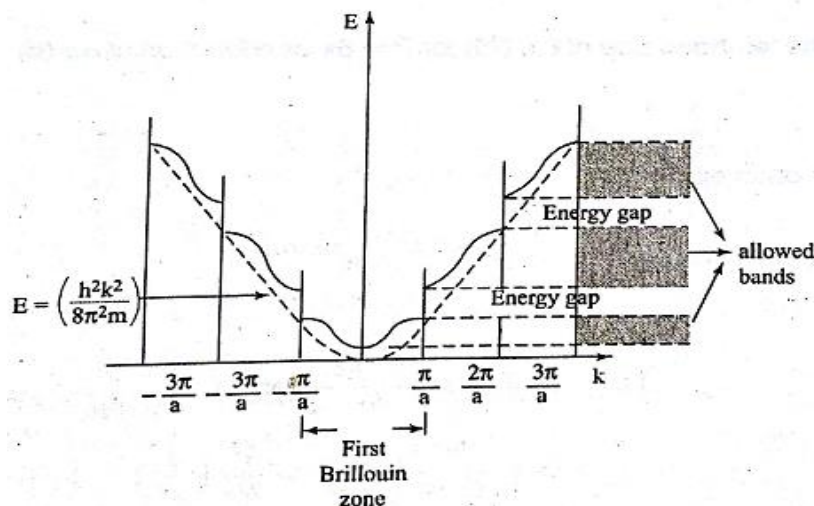
$$E = P^2 / 2 m$$

$$E = 1/2 m v^2 \text{ -----(14)}$$

The equation (11) shows all the electrons are completely free to move in the crystal without any constraints. Hence, no energy level exists i.e. all the energies are allowed to the electrons and shown in fig(5). This case supports the classical free electrons theory.

$$[(P/\alpha a) \sin \alpha a + \cos \alpha a], P \rightarrow 0$$

1.18 BRILLOUIN ZONE OR E-K DIAGRAM:



The Brillouin zone are the boundaries that are marked by the values of wave vector k , in which electrons can have allowed energy values. These represent the allowed values of k of the electrons in 1D, 2D, & 3D.

It is the energy spectrum of an electron moving in presence of a periodic potential field and is divided into allowed energy regions (allowed zones) or forbidden energy gaps (forbidden zones).

Allowed energy values lie in the region $k = -\pi/a$ to $+\pi/a$. This zone is called the first Brillouin zone. After a break in the energy values, called forbidden energy band, we have another allowed zone spread from $k = -\pi/a$ to $-2\pi/a$ and $+\pi/a$ to $+2\pi/a$. This zone is called the second Brillouin zone. Similarly, higher Brillouin zones are formed.

1.19 CONCEPT OF EFFECTIVE MASS OF ELECTRON:

When an electron in a periodic potential of lattice is accelerated by an known electric field or magnetic field, then the mass of the electron is called effective mass and is represented by m^*

To explain, let us consider an electron of charge 'e' and mass 'm' moving inside a crystal lattice of electric field E.

acceleration $a = eE / m$ is not a constant in the periodic lattice of the crystal. It can be considered that its variation is caused by the variation of electron's mass when it moves in the crystal lattice.

Therefore Acceleration $a = eE / m^*$

$$\text{Electrical force on the electron } F = m^* a \text{ -----(1)}$$

Applied Physics

I -B.Tech

Considering the free electron as a wave packet, the group velocity v_g corresponding to the particle's velocity can be written as

$$v_g = dw / dk = 2 \Pi dv / dk = (2 \Pi / h) dE / dk \text{ -----(2)}$$

where the energy $E = h \nu$ and $\hbar = h / 2 \Pi$.

$$\text{Acceleration } a = dv_g / dt = (1 / \hbar) d^2E / dk dt = (1 / \hbar) (d^2E / dk^2) dk / dt$$

$$\text{Since } \hbar k = p \text{ and } dp / dt = F,$$

$$dk / dt = F / \hbar$$

$$\text{Therefore } a = (1 / \hbar^2) (d^2E / dk^2) F$$

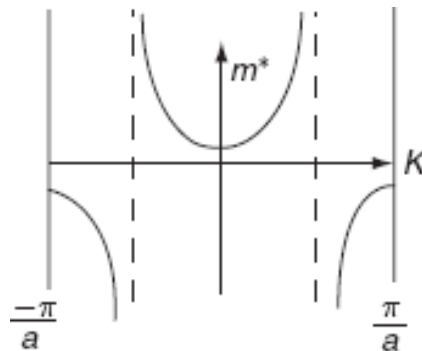
$$\text{Or } F = (\hbar^2 / (d^2E / dk^2)) a \text{ -----(3)}$$

Comparing eqns . (1) and (3) we get

$$m^* = \hbar^2 / (d^2E / dk^2) \text{ -----(4)}$$

This eqn indicates that the effective mass is determined by d^2E / dk^2

Variation of effective mass m^* with k : The graph shows variation of m^* with k . Near $k=0$, effective mass approaches m . As the k value increases m^* increases, reaching its maximum value known as infinite effective mass. Above the point of inflection, m^* is negative and as k tends to π/a , it decreases to a small negative value called negative effective mass in the lower region. The positively charged particle which can be located in the lower region called negative



1.20 ORIGIN OF ENERGY BAND FORMATION IN SOLIDS:

The band theory of solids explains the formation of energy bands and determines whether a solid is a conductor, semiconductor or insulator.

The existence of continuous bands of allowed energies can be understood starting with the atomic scale. The electrons of a single isolated atom occupy atomic orbitals, which form a discrete set of energy levels.

When two identical atoms are brought closer, the outermost orbits of these atoms overlap and interact. When the wave functions of the electrons of different atoms begin to overlap considerably, the energy levels corresponding to those wave functions split.

If more atoms are brought together more levels are formed and for a solid of N atoms, each of the energy levels of an atom splits into N energy levels. These energy levels are so close that they form an almost continuous band. The width of the band depends upon the degree of overlap of electrons of adjacent atoms and is largest for the outermost atomic electrons.

As a result of the finite width of the energy bands, gaps are essentially leftover between the bands called forbidden energy gap.

The electrons first occupy the lower energy levels (and are of no importance) then the electrons in the higher energy levels are of important to explain electrical properties of solids and these are called valence band and conduction band.

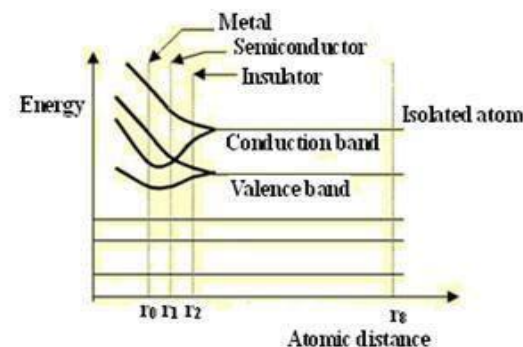
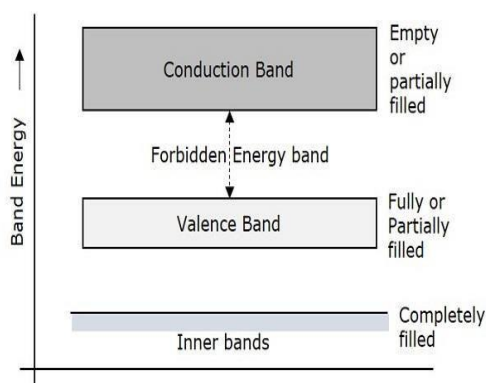


Fig: Formation of energy bands when atoms are closer

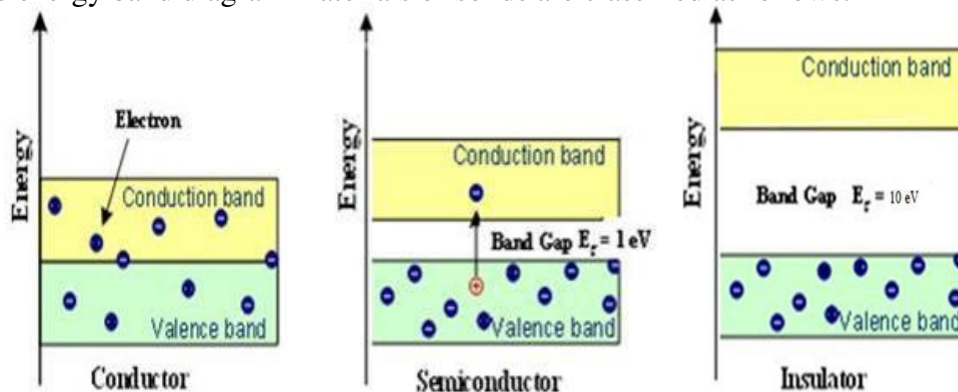
Valence band: A band occupied by valence electrons and is responsible for electrical, thermal and optical properties of solids and it is filled at 0K.

Conduction band: A band corresponding to outer most orbit is called conduction band and is the highest energy band and it is completely empty at 0K.

The **forbidden energy gap** between valence band conduction band is known as the energy band gap. By this solids are classified in to conductors, semiconductors and insulators.

1.21 CLASSIFICATION OF SOLIDS INTO CONDUCTORS , SEMICONDUCTORS & INSULATORS:

Based on the energy band diagram materials or solids are classified as follows:



Applied Physics

I -B.Tech

Conductors: In this kind of materials, there is no forbidden gap between the valence band and conduction band. It is observed that the valence band overlaps with the conduction band in metals as shown in figure. There are sufficient numbers of free electrons, available for electrical conduction and due to the overlapping of the two bands there is an easy transition of electrons from one band to another band takes place, and there no chance for the presence of holes. Resistivity of conductors is very small and it is very few milli ohm meters. ($\Omega \text{ m}$).

Examples: All metals (Na, Mg, Al, Cu, Ni Cu, Ag, Li, Ar etc)

Semiconductors: In semiconductors, there is a band gap exists between the valence band and conduction band and it is very less and it is the order of -1 to 2 eV are known as semiconductors. It will conduct electricity partially at normal conditions. The electrical resistivity values are 0.5 to 10^3 ohm meter. Due to thermal vibrations within the solid, some electrons gain enough energy to overcome the band gap (or barrier) and behave as conduction electrons. Conductivity exists here due to electronics and holes.

Examples: Silicon, Germanium, Ga As.

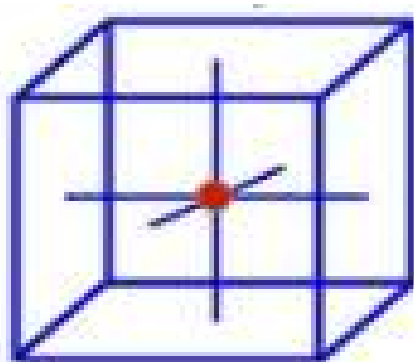
Insulators: In insulators, the width of forbidden energy gap between the valence band and conduction band is very large. Due to large energy gap, electrons cannot jump from V.B to C.B. Energy gap is of the order of ~ 10 eV and higher than semiconductors. Resistivity values of insulators are 10^7 to 10^{12} ohm-m. Electrons are tightly bound to the nucleus, no valence electrons are available.

Examples: Wood, rubber, glass.

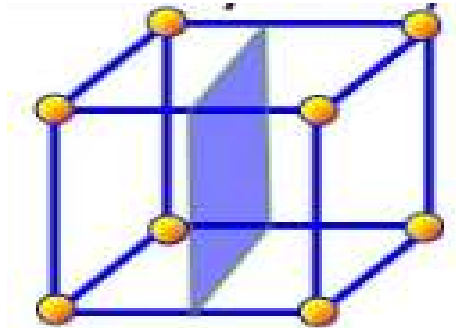
1.22 Symmetry in solids

A Symmetry operation is an operation that can be performed either physically or imaginatively that results in no change in the appearance of an object.

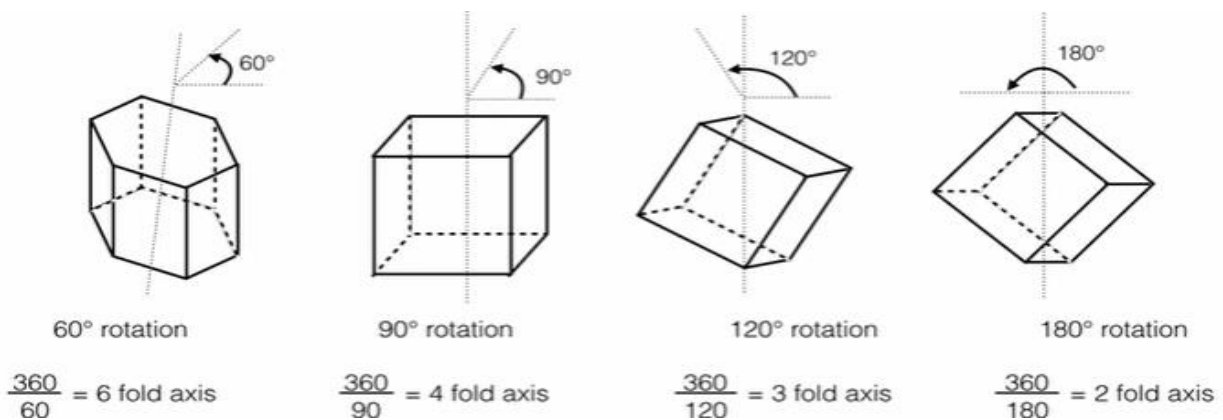
Center of symmetry: If an imaginary straight line can be passed through a crystal from any point on the surface of crystal such that the point of entry is similar to the point of exit, then the crystal has a center of symmetry.



Plane of symmetry: When an imaginary plane is passed through a crystal such that the portion of the crystal on one side of the plane is a reflection, or mirror image, of the portion on the other side of the plane, the plane is a **plane of symmetry** (often called a mirror plane).

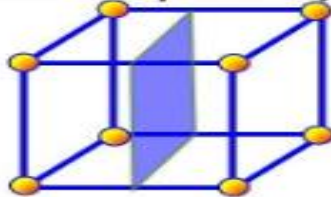


Axis of symmetry: When an imaginary line can be passed through a crystal such that the crystal can be rotated 360° about the line to fill the same space two, three, four, or six times, it has an **axis of symmetry**.

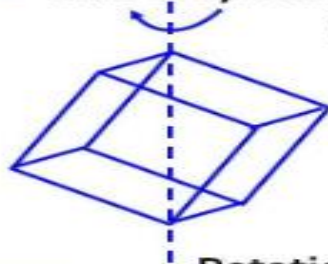


Symmetry Elements

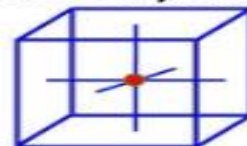
Plane of symmetry



Axis of symmetry

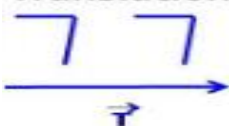


Centre of symmetry



Symmetry Operations

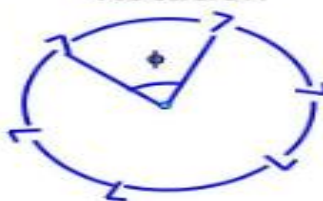
Translation



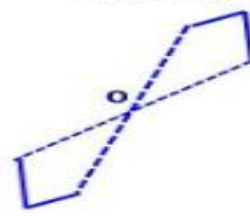
Reflection



Rotation



Inversion



Applied Physics

I -B.Tech

Short answer questions

1. What is a black body and explain its construction.
2. Explain about photoelectric effect.
3. Define Compton effect and Compton shift.
4. What are the postulates of Plank's hypothesis?
5. Distinguish between waves and particles.
6. What are matter waves? Explain their properties.
7. State and explain Heisenberg's uncertainty principle.
8. Explain Physical significance of wave function.
9. Write the drawbacks of classical free electron theory.
10. Discuss briefly about electron in periodic potential.
11. Write short note on origin of energy band formation in solids.
12. What are the possible symmetries in solids?

Essay questions

1. Derive expression for Plank's radiation law.
2. What is Compton effect? Derive Compton shift relation.
3. Describe experimental verification of matter waves using Davisson-Germer's experiment.
4. Derive time independent Schrodinger's wave equation for a free particle.
5. Derive an expression for energy and wave function of a particle bound to 1-D potential box.
6. Explain the Kronig-Penny model of solids and show that it leads to energy band structure.
7. Derive an expression for effective mass of an electron moving in energy bands of a solid.
8. Distinguish between conductor, semiconductors and insulators.

Applied Physics

I -B.Tech

Multiple answer questions

1. In the spectrum of black body radiation, Wein's law is applicable to []
 (A) Longer wavelength region (B) shorter wavelength region (C) any wavelength region (D) none of wavelengths
2. If ΔE and Δt are the uncertainties in the measurement of energy and time of a system, according to Heisenberg's uncertainty principle. []
 (A) $\Delta E \cdot \Delta t \geq h/4\pi$ (B) $\Delta E \cdot \Delta t \leq h/4\pi$ (C) $\Delta E \cdot \Delta t \geq h/2\pi$ (D) $\Delta E \cdot \Delta t \leq h/2\pi$
3. Matter waves are called []
 (A) Electromagnetic waves (B) de Broglie waves
 (C) ultrasonic waves (D) Planck waves
4. The wave function is []
 (A) The quantity associates with all waves
 (B) The probability amplitude of finding the particle in space
 (C) A mathematical function with four variables
 (D) A real function
5. Davisson and Germer experiment supports the concept ofof matter particles. []
 (A) Particle (B) Wave nature (C) Dual nature (D) Both B and C
6. The de Broglie wavelength is infinity when the speed of the matter particle is. []
 (A) Infinity (B) Finite (C) Zero (D) no relation
7. In the Davisson and Germer experiment, when 54 volts is applied to electrons, the first order diffraction spectrum is observed at the scattering angle of []
 (A) 25° (B) 50° (C) 36° (D) 42°
8. According toprinciple, it is not possible to know position and momentum of a particle at the same time. []
 (A) de-Broglie (B) Bragg (C) Schorindiger (D) Heisenberg
10. Schorindiger time independent wave equation is..... []
 (A) $\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} + \frac{8\pi^2(E-V)}{h^2} \varphi = 0$
 (B) $\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} + \frac{8m\pi^2(E-V)}{h^2} \varphi = 0$

Applied Physics

I -B.Tech

- (C) $\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} + \frac{8m\pi^2(E-V)}{\hbar^2} \varphi = 0$
- (D) $E\varphi = -\frac{\partial \varphi}{\partial t}$
11. Inside the one dimensional potential box the potential energy of the particle is []
 (A) Infinity (B) Depends on the size of box (C) Depends on the size of particle (D) Zero
12. The energy of the particle in one dimensional potential box is []
 (A) $E = \frac{n^2 \hbar^2}{8ml^2}$ (B) $E = \frac{n^2 \hbar^2}{8ml^2}$ (C) $E = \frac{n\hbar}{8ml^2}$ (D) $E = 0$
13. In which of the following de-Briglie wave equation is wrongly interpreted []
 (A) $\lambda = \frac{h}{mv}$ (B) $\lambda = \frac{h}{\sqrt{2mE}}$ (C) $\lambda = \frac{h}{p}$ (D) $\lambda = \frac{h}{mv^2}$
14. If an electron is moving under a field of V then the de-Broglie wave equation is given by []
 (A) $\lambda = \frac{12.2}{\sqrt{V}}$ (B) $\lambda = \frac{h}{\sqrt{2mV}}$ (C) $\lambda = \frac{h}{pV}$ (D) $\lambda = \frac{h}{mV}$
15. If an electron is moving under potential field of 100V the wave length is []
 (A) $12.2\text{\AA}$ (B) 12.2cm (C) $1.22\text{\AA}$ (D) 12.2m
16. Mater waves and electromagnetic waves are []
 (A) Same (B) Different (C) in some times both are similar (D) None.
17. Heisenberg's uncertainty principle is a consequence of []
 (A) de-Broglie (B) Bragg (C) Schorindiger (D) Davisson and Germer
18. A combination of position space and momentum space is called []
 (A) Macrostate (B) Microstate (C) Phase space (D) Whole space
19. Quantum theory successfully explains []
 a. Interference
 b. Polarization and blockbody radiation
 c. Photoelectric effect
 d. All
20. Dual nature of matter was proposed by []
 a. De Broglie
 b. Plnck
 c. Newton
 d. None
21. The target material in Davisson and Germer's Experiment []
 a. Ni,
 b. Au

Applied Physics

I -B.Tech

- c. Na
d. Cl
22. Diffraction angle for nickel in Davisson and Germer's Experiment []
(a).50°, (b).60°, (c).25°, (d).70°
23. Compton effect proves []
(a) light have particle nature, (b) light have wave nature, (c) a & b, (d) none of them
24. Davisson-Germer's experiment proves []
(a) light have particle nature, (b) light have wave nature, (c) a & b, (d) none of them
25. Matter waves are called []
(A)Electromagnetic waves (B)de Broglie waves
(C)ultrasonic waves (D)Planck waves